



High Availability and its Types



What is High Availability

- We live now in an IT based world, in which the success and the growth of the business depends mainly on the availability of the critical systems that run it.
- During Service Level Agreement (SLA) discussions, topics including Recovery Time Objective (RTO) and recovery Point Objective (RPO).
- That a second of downtime in financial or tracking systems will not be accepted.
- The availability of business applications depend also on the availability of the data that these applications process and store.
- What is the need for an application that is available without being able to reach to the databases that this application uses?
- provide a reliable high availability and disaster recovery solution that ensures the minimum possible downtime for the database systems in case of any crash, failure or disaster situation.
- **High Availability** means that the SQL Server instances or databases will be available and reachable, with the least possible downtime, in case of any server crash or failure.

What is High Availability (contd...)

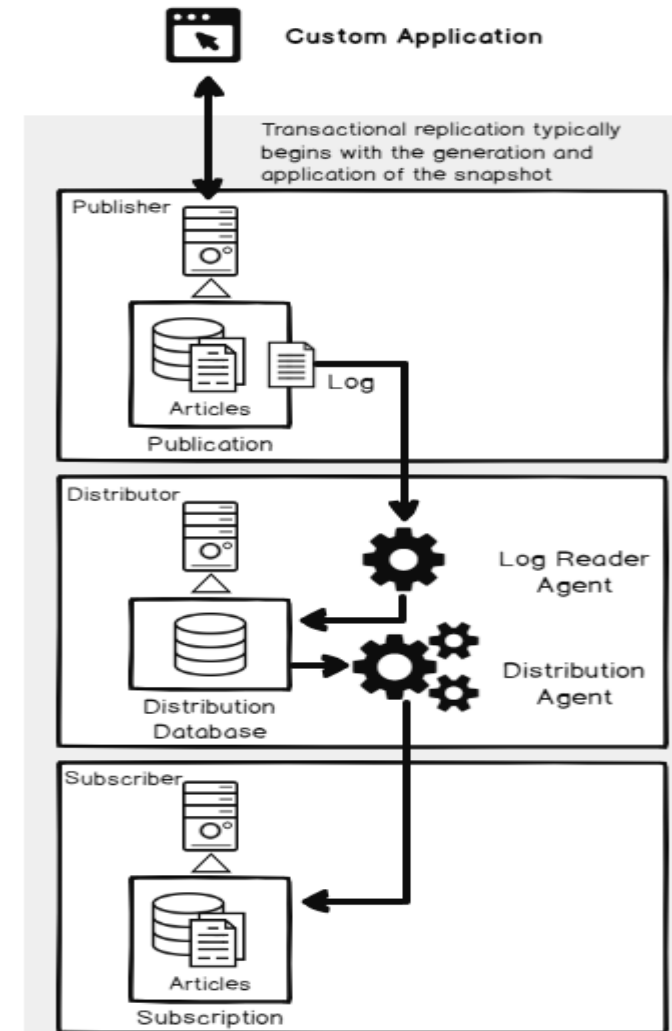
- The availability of the SQL Server that is provided by the solution is measured by the number of 9's in the availability percentage that this solution provides, and the total period of annual downtime, as per the Microsoft table below:

Number of 9's	Availability Percentage	Total Annual Downtime
2	99%	3 days, 15 hours
3	99.9%	8 hours, 45 minutes
4	99.99%	52 minutes, 34 seconds
5	99.999%	5 minutes, 15 seconds

- Disaster Recovery** means that, when the primary datacenter experiences a catastrophic event, such as an earthquake, flood or war, the system should be brought online and serve the users in another secondary site.

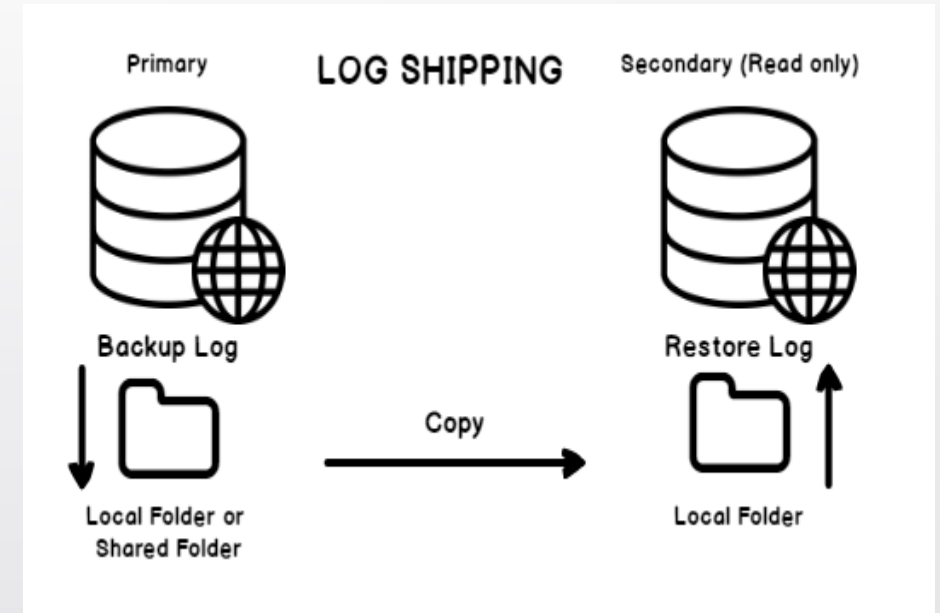
High Availability Types

- **Replication**
- The source data will be copied to destination through replication agents (jobs). Object level technology.
- Terminology
 - Publisher is source server.
 - Distributor is optional and stores replicated data for the subscriber.
 - Subscriber is the destination server.



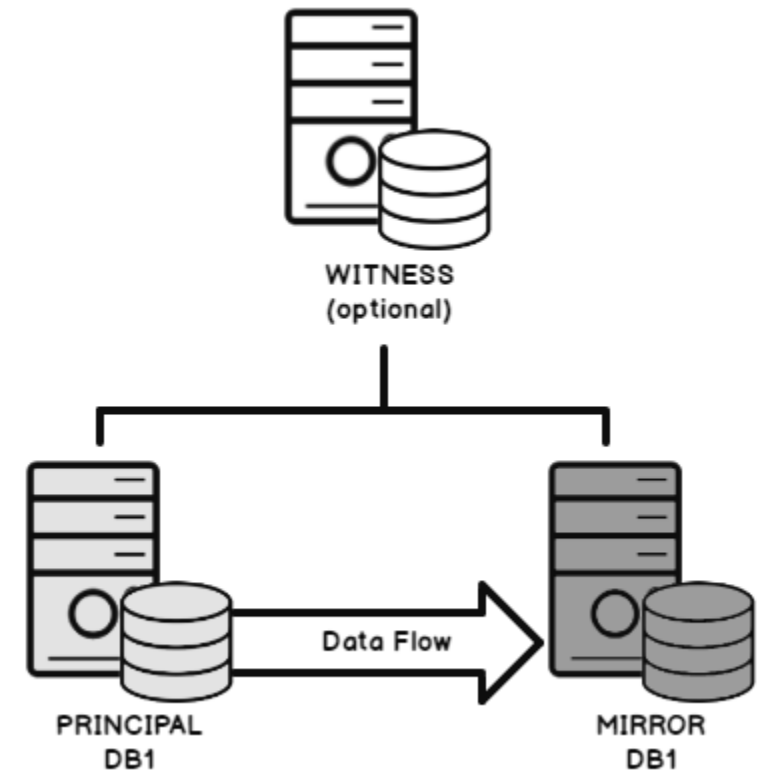
High Availability Types

- **Log Shipping**
- The source data will be copied to destination through Transaction Log backup jobs. Database level technology.
- Terminology
 - Primary server is source server.
 - Secondary server is destination server.
 - Monitor server is optional and will be monitored by log shipping status.



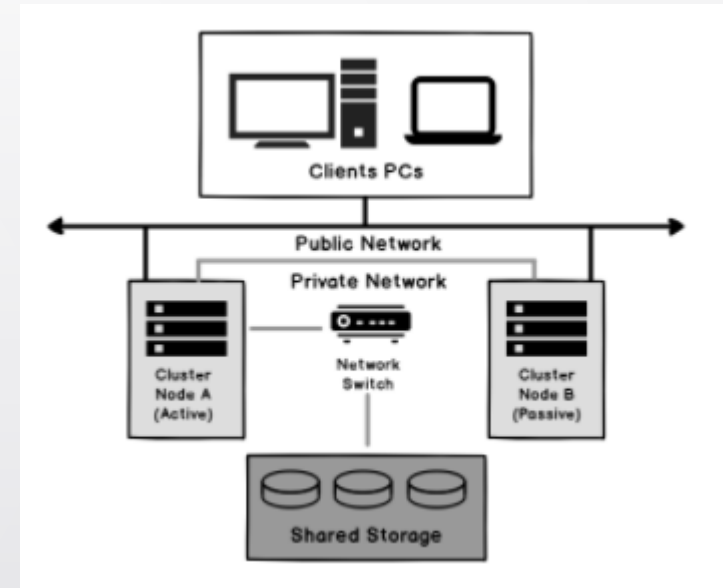
High Availability Types

- **Mirroring**
- The primary data will be copied to secondary through network transaction basis with the help of mirroring endpoint and port number. Database level technology.
- Terminology
 - Principal server is source server.
 - Mirror server is destination server.
 - Witness server is optional and used to make automatic failover.



High Availability Types

- **Clustering**
- The data will be stored in shared location which is used by both primary and secondary servers based on availability of the server. Instance level technology. Windows Clustering setup is required with shared storage.
- Terminology
 - Active node is where SQL Services are running.
 - Passive node is where SQL Services are not running.



High Availability Types

- **AlwaysON Availability Groups**
- The primary data will be copied to secondary through network transaction basis. Group of database level technology. Windows Clustering setup is required without shared storage.
- Terminology
 - Primary replica is source server.
 - Secondary replica is destination server.

